SHIWANK GUPTA

FPGA Developer | RTL Design & Verification Engineer

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Panchkula, India | Open to Relocation (Surat / Global) | Immediate Joiner

PROFESSIONAL SUMMARY

FPGA Developer and VLSI Design Engineer with complete FPGA design flow expertise — RTL coding (Verilog/SystemVerilog), synthesis, place & route, timing closure, testbench verification, and hardware board validation on Xilinx Vivado. Key results: 61.7% dynamic power reduction via hybrid clock gating (45 nm, 200 MHz); 69% PDP improvement in flip-flop benchmarking (28/65 nm). Hands-on UART/I2C/SPI protocol implementation. Immediate joiner, open to Surat relocation.

CORE TECHNICAL SKILLS

- **FPGA Tools:** Xilinx Vivado, ModelSim, Intel Quartus-compatible flows
- **HDL:** Verilog, SystemVerilog, Embedded C, Python, MATLAB
- **FPGA Flow:** RTL Design, Synthesis, Place & Route, Timing Closure, Bitstream Generation, Hardware Board Validation
- **Verification:** Self-Checking Testbenches, Directed Stress Testing, Multi-Corner Simulation (TT/SS/FF), DFT, Timing Closure
- **Protocols:** UART, I2C, SPI — hardware HDL implementation; AXI (conceptual)
- **EDA / ASIC:** Cadence Virtuoso, Synopsys VCS, PrimeTime PX, LTSpice, SPICE (BSIM4); STA, CTS, DRC/LVS
- **Low-Power:** Clock Gating (ICG), Power Gating, Multi-VTH, NTV Design
- **Platforms:** Cyclone V FPGA, Artix-7, Arduino, STM32, ESP32, Raspberry Pi

PROFESSIONAL EXPERIENCE

FPGA & RTL Design Engineer (Independent R&D) — *Jul 2025 – Present*

Nik-Coronics | Panchkula, India

- Built 14-project RTL-to-DV portfolio on Xilinx Vivado (Cyclone V/Artix-7): FSMs, FIFO, SRAM Memory Controller (altsyncram), UART Transmitter, 8-bit RISC Processor — full RTL to bitstream & hardware validation.
- Hybrid clock gating (ICG + dynamic) at RTL for 32-bit processor datapath: 61.7% dynamic power reduction (48.9 mW → 18.7 mW) at 45 nm, 200 MHz.
- Designed CMOS digital I/O pad with ESD protection; analyzed parasitic impact and signal integrity under high-frequency switching — relevant to high-speed serial interface challenges.
- Full-custom 6T SRAM cell in SPICE: SNM 92–108 mV across TT/SS/FF corners; flip-flop benchmarking at 28/65 nm CMOS — ATC-SCDFF achieved 0.32 fJ PDP (69% improvement).

ASIC Design Trainee — *Sep 2022 – Sep 2025*

TeamLease Services Limited | India

- 3-year structured program (60 hrs/month): digital ASIC design, RTL coding, logic synthesis, EDA-based verification, timing analysis; design reviews with senior engineers.

VLSI Design Intern — *Jul 2024 – Aug 2024*

Kurukshetra University Dept. of Electronic Science | Kurukshetra, India

- CMOS schematic, post-layout simulation, DRC/LVS in Cadence Virtuoso. Letter of Recommendation awarded.

Data Engineer Intern — *Mar 2025 – Jul 2025*

Samsung SDS India | Gurgaon, India

- Python-based data pipeline development within a global semiconductor-affiliated organization.

KEY FPGA & RTL PROJECTS

14-Day RTL-to-DV Architecture Portfolio GitHub — *Ongoing*

- FSMs, FIFO, SRAM Controller (Cyclone V altsyncram), UART Transmitter, 8-bit RISC Processor — 100% functional coverage via self-checking testbenches; hardware board validated on FPGA.

8-Bit RISC Processor — RTL to FPGA GitHub — *Oct–Nov 2025*

- Complete RISC architecture in Verilog/SystemVerilog; synthesized, placed & routed, timing closed, and deployed on FPGA hardware with comprehensive testbench.

Clock Gating Optimization for Dynamic Power Reduction GitHub — *Jan–Sep 2025*

- Static, dynamic & hybrid ICG on 32-bit RISC datapath, Verilog, 45 nm; 61.7% dynamic power reduction (48.9 mW → 18.7 mW) verified via Synopsys PrimeTime PX.

Low-Power Flip-Flop Architecture Benchmarking GitHub — *Jan 2024–Sep 2025*

- 4 architectures at 28 nm HKMG & 65 nm bulk CMOS (NTV); ATC-SCDFF: PDP 0.32 fJ — 69% improvement over baseline.

High-Speed Digital I/O Pad Design GitHub — *Dec 2025*

- CMOS pad with ESD protection; parasitic extraction & signal integrity analysis under high-frequency switching.

EV Regenerative Braking System GitHub — *Innovation Project*

- Hardware prototype for kinetic energy recovery during deceleration; demonstrated power electronics optimization.

EDUCATION

B.Tech — Electronics & Communication Engineering | UIET, Kurukshetra University | CGPA: 7.11 — *2022–2025*

Diploma — Electrical Engineering | Gold Medalist | Govt. Polytechnic College, Nanakpur | CGPA: 8.5 — *2019–2022*

CERTIFICATIONS

- Digital ASIC Design 3-Year Training — TeamLease Services (2022–2025)
- VLSI Design & Cadence Virtuoso — Kurukshetra University (Letter of Recommendation)
- Mandarin Chinese (Elementary) — Asia-Pacific semiconductor readiness